Impact of Recreational Marijuana Legalization on Crime: Evidence from California

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Abstract

This paper estimates the effect of California's 2016 legalization of recreational marijuana (Proposition 64) on local crime. Using 5,678 city-year observations from the FBI's Uniform Crime Reporting Program for 2013–2019, I compare California cities with cities in four states that permitted neither medical nor recreational marijuana during the period (Idaho, Kansas, South Carolina, and Wyoming) in a two-way fixed-effects difference-in-differences framework. Without covariates, legalization is associated with statistically significant increases in violent crime, property crime, burglary, and larceny-theft. These estimates fall substantially once time-varying controls for population, unemployment, and median household income are added. In the preferred specification, legalization has no statistically significant effect on violent crime (+1.80 per 100,000 residents; $p = 0.245$), aggravated assault, burglary, or motor vehicle theft, and there is only weak evidence of higher larceny-theft (+19.43; $p = 0.092$), equal to about 7 percent of California's pre-legalization mean. The results do not support claims that legalization produced a large increase in crime, although the modest property-crime estimates should be interpreted cautiously given the single treated state and concurrent criminal-justice reforms in California.

1. Introduction

Over the past decade, a growing number of U.S. states have moved from prohibiting marijuana to licensing and taxing its sale for recreational use. California, which in 1996 became the first state to legalize medical marijuana, legalized adult recreational use when voters approved Proposition 64 in November 2016. Because California is the most populous state in the country, its experience carries particular weight in the debate over whether legalization affects public safety.

Economic theory does not give a clear prediction about the direction of this effect. On one hand, a legal supply chain may displace illicit markets and the violence used to settle disputes within them, and police may redirect time previously spent on marijuana offenses toward more serious crime. On the other hand, legalization lowers the price and legal risk of consumption, which may increase intoxication-related offending, and cash-intensive retail outlets may create new targets for theft. Which of these channels dominates is ultimately an empirical question, and the existing evidence is mixed.

This paper estimates the effect of California's legalization on six categories of crime: violent crime, aggravated assault, property crime, burglary, larceny-theft, and motor vehicle theft. I construct a panel of 5,678 city-year observations from the FBI's Uniform Crime Reporting (UCR) Program for 2013–2019 and compare California cities with cities in Idaho, Kansas, South Carolina, and Wyoming, four states that allowed neither medical nor recreational marijuana during the sample period. The empirical strategy is a two-way fixed-effects difference-in-differences design that absorbs permanent differences across cities and shocks common to all cities in a given year. I report estimates with and without time-varying controls for population, the unemployment rate, and median household income.

Two main findings emerge. First, without covariates, legalization is associated with statistically significant increases of 4.83 violent crimes, 54.67 property crimes, 17.81 burglaries, and 38.25 larceny-thefts per 100,000 residents, along with a small decline in motor vehicle theft. Second, these estimates are not robust to controlling for local economic and demographic conditions. With covariates, the property crime estimate falls by about two-thirds, to 18.50 (p = 0.100); the violent crime, burglary, and motor vehicle theft estimates become small and statistically insignificant; and only larceny-theft retains weak statistical significance (19.43; p = 0.092), an increase of about 7 percent relative to California's pre-legalization mean. I interpret the

preferred estimates as evidence against large effects of legalization on crime, with suggestive but fragile evidence of a modest increase in larceny.

This paper contributes to a literature that has focused mainly on medical marijuana laws and on the earliest recreational adopters, Colorado, Washington, and Oregon. California differs from these states in size, in the maturity of its medical marijuana market, and in the scale of its licensed retail sector. The analysis also illustrates a methodological point that is relevant to single-state policy evaluations: unconditional difference-in-differences estimates can attribute to a policy what are in fact differential economic trends between the treated state and a small set of comparison states.

The rest of the paper proceeds as follows. Section 2 describes the policy background and related literature. Section 3 describes the data and presents descriptive evidence. Section 4 presents the empirical strategy, Section 5 reports the results, Section 6 discusses mechanisms and limitations, and Section 7 concludes.

2. Background and Related Literature

2.1 Marijuana Policy in California

California legalized medical marijuana through Proposition 215, the Compassionate Use Act, in 1996. Twenty years later, on November 8, 2016, voters approved Proposition 64, the Adult Use of Marijuana Act, with about 57 percent of the vote. The measure took effect the following day, allowing adults aged 21 and older to possess up to 28.5 grams of marijuana and to grow up to six plants at home. Commercial sales required a new state licensing system, and licensed recreational retailers did not open until January 1, 2018. Local governments retained the authority to prohibit commercial cannabis activity within their borders, so access to legal retail outlets varied considerably across cities.

The timing of the policy matters for interpretation. Legal possession began in late 2016, but the licensed retail market, the channel through which legalization is most likely to affect illicit markets and dispensary-related crime, operated only in 2018 and 2019 within the sample period. The estimates below therefore capture the combined short-run effect of legal possession and the first two years of retail sales.

California also adopted other criminal-justice reforms during this period. Proposition 47, approved in November 2014, reclassified several drug-possession and property offenses, including

theft of property valued at $950 or less, from felonies to misdemeanors. Proposition 57, approved in November 2016, expanded parole consideration for people convicted of nonviolent felonies. Because these reforms affected California but not the control states, they are potential confounders, a point I return to in Section 6.

2.2 Channels Linking Legalization and Crime

The literature identifies four main channels through which legalization could affect crime. The first is market displacement: legal supply may shrink illicit markets and the systemic violence associated with them. Consistent with this channel, Gavrilova et al. (2019) find that medical marijuana laws in states bordering Mexico reduced violent crime by about 13 percent. The second is the reallocation of police resources. If officers spend less time on marijuana offenses, they may solve or deter more serious crimes; Makin et al. (2019) find that clearance rates for some offenses improved in Colorado and Washington after legalization.

The third channel is consumption. Legalization increases marijuana use among adults, and opponents argue that greater use could raise crime directly through intoxication or indirectly by serving as a gateway to harder drugs. Sabia et al. (2024) find that recreational marijuana laws increase adult marijuana use but find no evidence that they increase harder drug use or arrests for violent and property crime. The fourth channel operates through retail outlets. Dispensaries handle large amounts of cash and product, which could make them targets for theft, but they also invest heavily in security and generate foot traffic. The causal evidence favors the second effect: Chang and Jacobson (2017) find that crime rose around Los Angeles medical marijuana dispensaries ordered to close relative to those allowed to remain open, and Brinkman and Mok-Lamme (2019) find that dispensary openings reduced neighborhood crime in Denver.

2.3 Empirical Evidence

Studies of medical marijuana laws generally find no increase in crime. Using state panel data for 1990-2006, Morris et al. (2014) find no crime-increasing effect of medical marijuana laws and some evidence of lower homicide and assault rates. Huber et al. (2016) estimate that medical marijuana legalization reduced robberies, larcenies, and burglaries by 4 to 12 percent. Chu and Townsend (2019) find no effect of medical marijuana laws on violent or property crime at the national level, but estimate reductions of about 20 percent in both categories in California, the only state in which they find a strong effect.

Evidence on recreational legalization is more mixed. Dragone et al. (2019) compare counties on either side of the Washington-Oregon border before Oregon legalized and find that legalization in Washington reduced rapes and thefts. In contrast, Wu et al. (2021), using a design close to the one in this paper — Oregon counties compared with counties in 19 states without legal marijuana — find that legalization increased both violent and property crime and that the start of retail sales was associated with higher property crime. Using variation across all adopting states, Sabia et al. (2024) find no effect on violent or property crime arrests. Anderson and Rees (2023) review this literature more broadly. Taken together, these findings suggest that effects may depend on the state, the geographic unit of analysis, and how long after legalization outcomes are measured.

3. Data

3.1 Data Sources

Crime data come from the FBI's UCR Program, which compiles offenses known to law enforcement as reported by local agencies. I use annual city-level data for 2013–2019 and express each outcome as offenses per 100,000 residents. The six outcomes are violent crime, which comprises murder and nonnegligent manslaughter, rape, robbery, and aggravated assault; aggravated assault, the largest component of violent crime; property crime, which comprises burglary, larceny-theft, and motor vehicle theft; and each of these three property offenses separately. Disaggregating outcomes allows the analysis to distinguish offenses that might respond to changes in illicit-market activity from crimes of opportunity such as larceny.

I use three time-varying covariates. City population, from the UCR, accounts for changes in city size. The unemployment rate, from the Bureau of Labor Statistics, and median household income for householders aged 25 to 44, from the U.S. Census Bureau, capture local economic conditions, which economic models of crime predict will affect the relative returns to legal and illegal activity (Becker, 1968). Appendix Table A1 lists all variable definitions and sources.

3.2 Treatment and Control Groups

The treatment group consists of cities in California. The control group consists of cities in Idaho, Kansas, South Carolina, and Wyoming, none of which permitted medical or recreational

marijuana at any point during 2013-2019.[1] Using never-treated states avoids contaminating the comparison with states whose own marijuana policies changed during the period. The trade-off is that the control states are smaller and more rural than California, and crime levels differ between the two groups (Table 1). The difference-in-differences design does not require similar levels of crime, but it does require that crime would have followed similar trends in the absence of legalization, an assumption discussed in Section 4.

The pre-legalization period is 2013–2015 and the post-legalization period is 2016–2019. Because Proposition 64 took effect only in November 2016, most of 2016 precedes legalization. Coding 2016 as a post-legalization year therefore places roughly ten months of pre-policy data in the post period, which should, if anything, bias the estimates toward zero. The final sample contains 5,678 city-year observations; on average, about 460 California cities and 350 control-state cities report in each year.

3.3 Descriptive Statistics

Table 1 reports summary statistics for the full sample, California cities, and control-state cities, both over the full period and separately for the pre- and post-legalization periods. Two features stand out. First, crime rates are higher on average in California than in the control states, particularly for property crime (430.53 versus 277.46 per 100,000 residents over 2013–2019) and motor vehicle theft (71.35 versus 16.52). Second, the California distributions are extremely dispersed. The standard deviation of the California property crime rate is more than six times its mean, compared with less than one times the mean in the control states, which indicates a highly right-skewed distribution in which a small number of California cities have very high per-capita rates. This feature motivates the robustness checks proposed in Section 6.

[1]Some of these states enacted narrow laws allowing patients with specific conditions, such as severe epilepsy, to use low-THC cannabidiol (CBD) products. These laws do not create a legal market for marijuana and are not generally classified as medical marijuana laws.

Table 1. Summary Statistics: Crime Rates by Group and Period

	Full sample			California cities			Control-state cities		
	N	Mean	SD	N	Mean	SD	N	Mean	SD
Panel A. Full period, 2013–2019									
Violent crime	5,678	43.47	156.61	3,224	47.69	204.90	2,454	37.93	39.34
Aggravated assault	5,678	28.31	74.81	3,224	28.25	95.41	2,454	28.40	31.51
Property crime	5,645	364.86	2,151.74	3,223	430.53	2,839.83	2,422	277.46	217.89
Burglary	5,654	68.29	330.90	3,224	78.07	435.69	2,430	55.31	51.44
Larceny-theft	5,668	248.76	1,421.34	3,223	281.08	1,878.39	2,445	206.16	172.21
Motor vehicle theft	5,675	47.67	445.74	3,224	71.35	590.13	2,451	16.52	17.41
Panel B. Pre-legalization, 2013–2015									
Violent crime	2,440	41.72	134.63	1,384	45.03	175.70	1,056	37.39	37.40
Aggravated assault	2,440	27.95	71.65	1,384	28.00	91.64	1,056	27.89	29.37
Property crime	2,432	371.70	1,979.50	1,384	426.29	2,615.20	1,048	299.61	234.18
Burglary	2,434	71.43	233.75	1,384	77.24	306.13	1,050	63.78	55.40
Larceny-theft	2,438	252.63	1,323.53	1,384	277.08	1,749.07	1,054	220.54	185.20
Motor vehicle theft	2,438	47.68	442.62	1,384	71.97	586.17	1,054	15.78	18.20
Panel C. Post-legalization, 2016–2019									
Violent crime	3,238	44.79	171.33	1,840	49.69	224.39	1,398	38.34	40.75
Aggravated assault	3,238	28.58	77.12	1,840	28.44	98.17	1,398	28.78	33.05
Property crime	3,213	359.68	2,273.74	1,839	433.73	2,998.51	1,374	260.57	203.08
Burglary	3,220	65.91	388.55	1,840	78.69	512.06	1,380	48.87	47.22
Larceny-theft	3,230	245.84	1,491.13	1,839	284.09	1,970.59	1,391	195.27	160.88
Motor vehicle theft	3,237	47.66	448.14	1,840	70.89	593.24	1,397	17.07	16.78

Notes: The unit of observation is the city-year. Crime rates are offenses known to law enforcement per 100,000 residents, from the FBI Uniform Crime Reporting (UCR) Program. Violent crime comprises murder and nonnegligent manslaughter, rape, robbery, and aggravated assault; property crime comprises burglary, larceny-theft, and motor vehicle theft. Control-state cities are located in Idaho, Kansas, South Carolina, and Wyoming, none of which permitted medical or recreational marijuana during 2013–2019. N varies across rows because some agencies did not report every offense category in every year.

Table 2 converts the period means into a simple two-by-two difference-in-differences comparison. In the control states, property crime fell by 39.04 offenses per 100,000 residents between the two periods, driven by declines in larceny-theft (-25.27) and burglary (-14.91). This decline is consistent with the broader national decline in property crime over the period (Federal Bureau of Investigation, 2019). Over the same period, property crime in California cities rose slightly (+7.44). The resulting unadjusted difference-in-differences for property crime is +46.48, with the largest contributions from larceny-theft (+32.28) and burglary (+16.36). Violent crime rose in both groups, by 4.66 in California and 0.95 in the control states, for an unadjusted difference of +3.71. Aggravated assault and motor vehicle theft changed little in either group.

Two points about Table 2 are worth emphasizing. The relative increase in California property crime reflects falling crime in the control states more than rising crime in California, so the comparison depends heavily on whether those declines represent what would have happened in California without legalization. In addition, these comparisons do not account for differences in local economic conditions and should be read as descriptive. The regression analysis below examines how much of each gap survives the inclusion of fixed effects and covariates.

Table 2. Unadjusted Difference-in-Differences in Mean Crime Rates

	California cities			Control-state cities			
	2013–2015	2016–2019	Change	2013–2015	2016–2019	Change	Difference
	(1)	(2)	(3)	(4)	(5)	(6)	(3) − (6)
Violent crime	45.03	49.69	+4.66	37.39	38.34	+0.95	+3.71
Aggravated assault	28.00	28.44	+0.44	27.89	28.78	+0.89	−0.45
Property crime	426.29	433.73	+7.44	299.61	260.57	−39.04	+46.48
Burglary	77.24	78.69	+1.45	63.78	48.87	−14.91	+16.36
Larceny-theft	277.08	284.09	+7.01	220.54	195.27	−25.27	+32.28
Motor vehicle theft	71.97	70.89	−1.08	15.78	17.07	+1.29	−2.37

Notes: Entries are mean crime rates per 100,000 residents from Panels B and C of Table 1. Columns (3) and (6) report the change from 2013–2015 to 2016–2019 within each group, and the final column reports the difference between these changes. These comparisons do not adjust for city fixed effects, year fixed effects, or covariates; regression-adjusted estimates appear in Table 3.

4. Empirical Strategy

I estimate the effect of legalization using the two-way fixed-effects specification

$$Y_{ist} = \delta(CA_s \times Post_t) + X'_{ist}\beta + \lambda_i + \theta_t + \varepsilon_{ist} \tag{1}$$

where Y_{ist} is the crime rate in city i in state s in year t; CA_s is an indicator for cities in California; $Post_t$ is an indicator for the years 2016–2019; X_{ist} is a vector of time-varying covariates (population, the unemployment rate, and median household income); λ_i and θ_t are city and year fixed effects; and ε_{ist} is the error term. The city fixed effects absorb all time-invariant differences across cities, including the average difference between California and the control states, and the year fixed effects absorb shocks common to all cities in a given year, including the average difference between the pre- and post-legalization periods. The main effects of CA_s and $Post_t$ are therefore collinear with the fixed effects and are not separately identified.[2] The coefficient of interest, δ,

[2]A regression that includes the California and post-period indicators alongside city and year fixed effects omits these terms automatically because of perfect collinearity; the estimate of δ is unaffected.

measures the average change in crime in California cities after legalization relative to the change in control-state cities over the same period (Meyer, 1995; Angrist & Pischke, 2009).

Identification. The estimate of δ has a causal interpretation under the parallel trends assumption: conditional on the fixed effects and covariates, crime in California cities would have followed the same trajectory as crime in control-state cities had California not legalized recreational marijuana. The assumption also requires that no other California-specific shock coinciding with the post-legalization period affected crime. I estimate equation (1) with and without X_{ist}. The specification without covariates relies on unconditional parallel trends. The specification with covariates allows California and the control states to follow different trends to the extent that those trends are explained by population and local economic conditions. I treat the covariate-adjusted specification as preferred, with one caveat: if legalization itself affected the covariates, for example by raising local employment or income through the new cannabis industry, controlling for them would absorb part of the policy effect (Angrist & Pischke, 2009).

Inference. Standard errors are clustered at the city level, which allows for arbitrary correlation in the errors of a given city over time (Bertrand et al., 2004). However, legalization varies only at the state level, and only one state is treated. In this setting, conventional clustered standard errors can understate uncertainty because they do not account for state-level shocks common to all California cities (Conley & Taber, 2011). The reported *p*-values are therefore likely to overstate precision, and estimates of marginal significance should be interpreted with particular caution.

5. Results

5.1 Estimates without Covariates

Panel A of Table 3 reports estimates of equation (1) with city and year fixed effects only. Legalization is associated with statistically significant increases of 4.83 violent crimes, 54.67 property crimes, 17.81 burglaries, and 38.25 larceny-thefts per 100,000 residents, and with a statistically significant decline of 1.94 motor vehicle thefts. The aggravated assault estimate is small and insignificant (0.36; $p = 0.556$). These estimates are similar in sign and magnitude to the unadjusted differences in Table 2 for every outcome except aggravated assault, for which both are close to zero. The fixed effects alone therefore do little to change the comparison: the baseline

estimates are driven largely by the decline in property crime in the control states that did not occur in California.

Table 3. Effect of Recreational Marijuana Legalization on Crime in California

	Violent offenses		Property offenses			
	Violent crime	Aggravated assault	Property crime	Burglary	Larceny-theft	Motor vehicle theft
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. City and year fixed effects only						
California × Post	4.83***	0.36	54.67**	17.81***	38.25**	−1.94***
	[0.000]	[0.556]	[0.014]	[0.004]	[0.017]	[0.000]
Panel B. Adding population, unemployment rate, and median household income						
California × Post	1.80	0.61	18.50	0.52	19.43*	−0.20
	[0.245]	[0.570]	[0.100]	[0.690]	[0.092]	[0.850]
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,678	5,678	5,678	5,678	5,678	5,678
California mean, 2013–2015	45.03	28.00	426.29	77.24	277.08	71.97

Notes: Each coefficient comes from a separate estimate of equation (1) and is the coefficient on the interaction between an indicator for California cities and an indicator for 2016–2019. Dependent variables are offenses per 100,000 residents. The control group consists of cities in Idaho, Kansas, South Carolina, and Wyoming; the sample period is 2013–2019. Panel B adds city population, the unemployment rate, and median household income for householders aged 25 to 44 as time-varying controls. *p*-values based on standard errors clustered at the city level are reported in brackets. Because treatment varies only at the state level and a single state is treated, these *p*-values likely overstate precision (see Section 4). The final row reports the mean of each dependent variable for California cities before legalization. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Estimates with Covariates

Panel B adds population, the unemployment rate, and median household income. The estimates change substantially. The violent crime estimate falls from 4.83 to 1.80 and is no longer statistically significant ($p = 0.245$), and the aggravated assault estimate remains small and insignificant (0.61; $p = 0.570$). Among property offenses, the property crime estimate falls by about two-thirds, from 54.67 to 18.50, with a *p*-value of 0.100, at the boundary of the conventional 10 percent threshold. The burglary estimate nearly disappears (0.52; $p = 0.690$), and the motor vehicle theft estimate shrinks to −0.20 ($p = 0.850$). Larceny-theft is the only outcome with an estimate that is statistically significant at the 10 percent level: legalization is associated with 19.43 additional larceny-thefts per 100,000 residents ($p = 0.092$), about half the baseline estimate.

5.3 Magnitudes

To gauge economic significance, the final row of Table 3 reports the mean of each outcome in California before legalization. The preferred violent crime and property crime estimates each equal about 4 percent of the corresponding pre-legalization mean, and the larceny-theft estimate equals about 7 percent. Even taken at face value, these effects are modest, and only the larceny-theft estimate is distinguishable from zero at conventional levels.

5.4 Why Do the Covariates Matter?

Adding three covariates changes the estimates this much only if those covariates evolved differently in California than in the control states and are themselves correlated with crime. In the baseline specification, these differential trends load onto the legalization indicator. Controlling for them shifts the identifying assumption from unconditional to conditional parallel trends, which is more plausible given how different the treatment and control states are. At the same time, the size of the change is a warning sign. If three observable covariates move the estimates this much, other unobserved differences between California and the control states may matter as well, which reinforces the need for the additional checks described in Section 6.

6. Discussion

6.1 Interpreting the Results

The absence of a significant effect on violent crime in the preferred specification is consistent with multistate evidence on recreational marijuana laws (Sabia et al., 2024) and with most studies of medical marijuana laws (Morris et al., 2014; Chu & Townsend, 2019). It differs from the findings of Wu et al. (2021) for Oregon, which may reflect differences between the two states, in the geographic unit of analysis (counties versus cities), or in the set of comparison states.

The weakly significant larceny-theft estimate is in the same direction as the property crime increase that Wu et al. (2021) document in Oregon, and several mechanisms could produce it. Licensed retail outlets could create new targets for theft, although the best available causal evidence suggests that dispensaries reduce rather than increase crime in their immediate vicinity (Chang & Jacobson, 2017; Brinkman & Mok-Lamme, 2019). Changes in reporting or enforcement after a major policy change could also raise recorded crime without any change in offending. Finally, the estimate may partly reflect California's other criminal-justice reforms. Proposition 47 reduced penalties for theft of property worth $950 or less beginning in November 2014, and Bartos

and Kubrin (2018) find suggestive, though not robust, evidence that larceny rose in California after its passage. If the effects of Proposition 47 grew over time, part of that increase would fall in the post-legalization period and would be attributed to legalization in equation (1). The current design cannot separate these explanations.

6.2 Limitations and Extensions

Several limitations suggest directions for future work. First, with one treated state and four control states, inference is fragile (Conley & Taber, 2011), and the results may depend on the choice of comparison states. Synthetic control methods (Abadie et al., 2010), which construct a weighted comparison unit from a larger pool of states, and randomization inference based on placebo assignments of treatment to the control states would provide more credible tests.

Second, the analysis does not formally test the parallel trends assumption. With three pre-legalization years, an event-study specification that interacts the California indicator with each year of the sample would show whether crime trends in California and the control states diverged before 2016. The same specification would also trace out how any effects evolved between the period of legal possession only (2016–2017) and the period of licensed retail sales (2018–2019).

Third, redefining the post-legalization period to begin in 2017, or estimating a separate effect for the retail period beginning in 2018, would align the treatment definition more closely with the policy's actual timing. Fourth, because the California distribution of crime rates is highly skewed, estimating population-weighted models, winsorizing extreme rates, or using log rates or count models would show whether the results are driven by a small number of cities. Finally, participation in the UCR Program is voluntary and reporting practices vary across agencies and over time, so recorded crime may change for reasons unrelated to underlying offending.

7. Conclusion

This paper uses city-level UCR data for 2013–2019 and a two-way fixed-effects difference-in-differences design to estimate the effect of California's recreational marijuana legalization on crime, comparing California cities with cities in four states that did not permit medical or recreational marijuana. Estimates without covariates suggest that legalization increased violent and property crime, but these estimates are largely explained by differences in local economic and demographic conditions between California and the control states. In the preferred specification, legalization has no statistically significant effect on violent crime, aggravated assault, burglary, or

motor vehicle theft, and there is only weak evidence of an increase in larceny-theft of about 7 percent relative to the pre-legalization mean.

For policymakers, these results do not support the claim that legalization led to a large increase in crime in California, at least in the first years after Proposition 64. The larceny-theft estimate is modest and fragile, and it overlaps in time with other reforms that also affected theft. Monitoring property crime as legal retail markets mature, and separating the effects of legalization from those of concurrent criminal-justice reforms, are important next steps for research.

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Appendix

Table A1. Variable Definitions and Data Sources

Variable	*Definition*	*Source*
Violent crime rate	Murder and nonnegligent manslaughter, rape, robbery, and aggravated assault per 100,000 residents	FBI Uniform Crime Reporting Program
Aggravated assault rate	Aggravated assaults per 100,000 residents	FBI Uniform Crime Reporting Program
Property crime rate	Burglary, larceny-theft, and motor vehicle theft per 100,000 residents	FBI Uniform Crime Reporting Program
Burglary rate	Burglaries per 100,000 residents	FBI Uniform Crime Reporting Program
Larceny-theft rate	Larceny-thefts per 100,000 residents	FBI Uniform Crime Reporting Program
Motor vehicle theft rate	Motor vehicle thefts per 100,000 residents	FBI Uniform Crime Reporting Program
California × Post	Equals 1 for California cities in 2016–2019 and 0 otherwise	Constructed
Population	Resident population of the city	FBI Uniform Crime Reporting Program
Unemployment rate	Unemployment rate, in percent	U.S. Bureau of Labor Statistics
Median household income	Median household income for householders aged 25 to 44, in dollars	U.S. Census Bureau

Notes: All variables are measured at the city-year level for 2013–2019.